

ADAPTIVE PATCH BASED CONVOLUTIONAL NEURAL NETWORK FOR ROBUST DEHAZING

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ABSTRACT

We present a novel deep learning-based dehazing method using adaptive patch splits. Our method applies quad-tree decomposition to an input image, yielding multiple patches with adaptive sizes. Then, each patch is fed into a Convolutional Neural Network (CNN) and classified into a single transmission value, in which a transmission map comprises transmission values from all patches. Homogeneous regions in the image are typically decomposed into large patches. Thus the method can save computational cost. Non-homogeneous regions are divided into small patches, which helps preserve local details in a transmission map. To train CNN, we synthesize numerous hazy images from haze-free images. Experimental results demonstrate our method surpasses state-of-the-art deep learning based algorithms quantitatively and qualitatively.

Index Terms— Dehazing, Adaptive Patch Classification, Quad-tree Decomposition

1. INTRODUCTION

The performance of computer vision algorithms depends highly on the environments from which the images are captured. Hazy images captured in unfavorable weather usually have low contrast, color fidelity and visibility. Degraded images negatively affect the performance of computer vision algorithms because most algorithms assume haze-free images as inputs. For example, hazy input images can cause erroneous detection or decision for autonomous vehicles and video surveillance systems. In these images, dehazing is necessary to improve the accuracy of the systems. However, several challenges still remain in dehazing, due to the inherent ambiguity in the separation of the haze and from the underlying scene.

In this paper, we present a robust haze-removal algorithm using deep learning based patch classification. To begin with, the input image is divided into non-overlapping patches with adaptive size, as shown in Fig. 1. For this, we adopt quad-tree decomposition techniques. Given the adaptive patches, Convolutional Neural Network (CNN) estimates the transmission value of each patch. A single transmission map is constructed by gathering transmission values of all patches. Then, a haze-

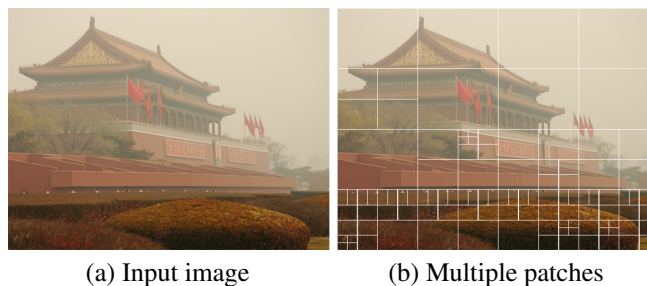


Fig. 1. Basic idea of our method. By quad-tree decomposition, an input image is divided into multiple patches. The small patches preserve the local details of non-homogeneous regions. Thus, CNN can estimate accurate transmission values in these patches. By dividing the image into large patches in homogeneous regions, our method can reduce the total number of patches and save computational cost.

free image is obtained from the transmission map and the estimated atmospheric light. Fig. 2 illustrates a pipeline of the proposed algorithm.

The main contributions of our approach are three-fold:

- We present a novel patch-classification based dehazing algorithm. To the best of our knowledge, our method is the first to use a deep learning method for patch classification in dehazing problems.
- We adaptively decompose the input image into multiple patches by using quad-tree decomposition. Our method trades off dehazing accuracy against computation cost by using adaptive patches.
- We improve the accuracy of the atmospheric light estimation by recycling the patch classification results.

2. RELATED WORK

For dehazing, additional information such as geographic information and near-infrared images is used [1, 2, 3]. He *et al.* [4] propose a dark channel prior that enables to obtain haze-free images with single images. They discover that the minimum values in local regions (except sky regions) of haze-free images are close to zero in at least one color channel.

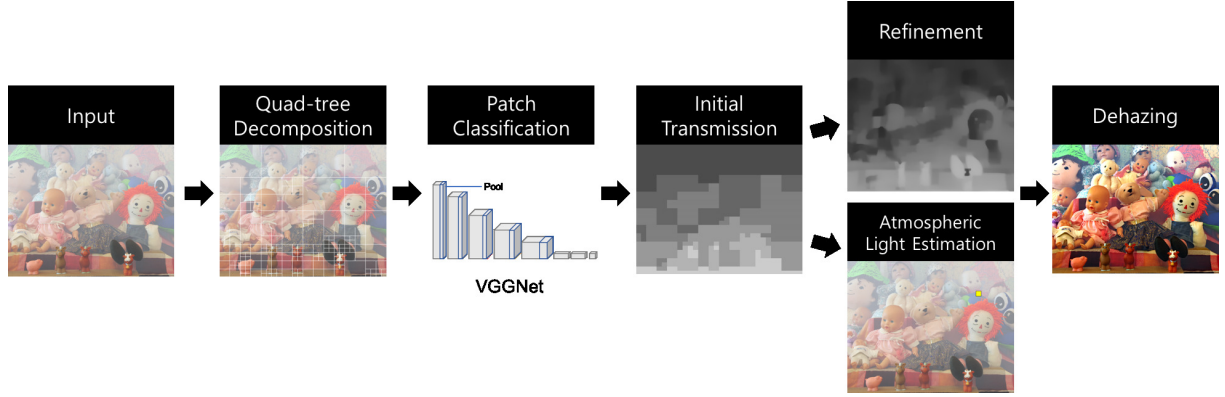


Fig. 2. Overall procedure of the proposed algorithm. The yellow box indicates the pixel, of which value is used for the atmospheric light estimation.

Tang *et al.* [5] present a regression model based on random forests using haze relevant features to improve visibility of images. In this paper, we apply a deep learning based algorithm to implement the aforementioned ideas.

Recently, CNN has attracted increasing interest in single image haze removal [6, 7, 8]. For example, MSCNN [6] proposes a multi-scale CNN network that passes from coarse to fine-scale networks. DehazeNet [7] estimates a transmission map using a CNN network that consists of feature extraction, multi-scale mapping, local extremes, and nonlinear regression. AOD-Net [8] proposes an end-to-end trainable dehazing model based on a reformulated atmospheric scattering model. These methods can make halo effects or are unable to preserve local details in certain images.

There are a few algorithms that adopt quad-tree to haze removal problems [9, 10]. Kim *et al.* [9] use quad-tree to estimate atmospheric light by recursively dividing an input image into four sub-blocks and select the brightest block with a maximum average. By using the quad-tree decomposition, Baig *et al.* [10] exclude bright objects for accurate estimation of atmospheric light. Unlike these methods, we make full use of the quad-tree decomposition to configure the whole framework (*i.e.*, to estimate both transmission and atmospheric light).

3. DEEP LEARNING BASED PATCH CLASSIFICATION FOR ROBUST DEHAZING

In this section, we describe the patch classification approach in detail. First, we introduce an atmospheric scattering model to explain hazy images physically. McCartney [11] proposes the model:

$$I(\mathbf{x}) = J(\mathbf{x})t(\mathbf{x}) + A(1 - t(\mathbf{x})), \quad (1)$$

where $I(\mathbf{x})$ is the observed hazy image, $J(\mathbf{x})$ the haze-free image to be recovered, $t(\mathbf{x})$ the transmission, A the global atmospheric light, and $\mathbf{x}$ a pixel coordinate. The transmission map $t(\mathbf{x})$ describes the portion of light that can reach

the camera without scattering. When the atmosphere is homogeneous, $t(\mathbf{x})$ can be modeled as $e^{-\beta d(\mathbf{x})}$, where β is the scattering coefficient of the atmosphere and d is the distance between scene points and the camera. The haze model in (1) can be written as

$$J(\mathbf{x}) = \frac{I(\mathbf{x}) - A}{t(\mathbf{x})} + A. \quad (2)$$

As shown in (2), we need to accurately estimate the transmission map $t(\mathbf{x})$ and the atmospheric light A to recover the haze-free image $J(\mathbf{x})$ from $I(\mathbf{x})$.

3.1. Initial transmission map using patch classification

It is not trivial to collect training data in supervised learning for dehazing problems. It is impossible to have pairs of hazy and haze-free images for every object in the scene. Therefore, many dehazing methods synthesize hazy images from natural images based on (1). Our basic idea stems from Tang's method [5]. Tang *et al.* make synthesized hazy patches for training, in which each local patch is represented by a single transmission value based on the assumption that depth is locally constant. They generate handcrafted features (*i.e.*, haze relevant features) and classify them with Random Forests. In our method, good feature maps for dehazing are automatically trained and classified into one of the transmission values by deep learning. In particular, we enforce $t(\mathbf{x})$ to range from 0.1 to 1.0 and fix A to 1 for synthesizing hazy patches using (1). For training, synthesized training images are fed into VGGNet [12] and classified into 10 classes. The classes have values that ranges from 0 to 1.0 at every interval of 0.1. For testing, we divide an input image into non-overlap patches, perform patch classification, and label each patch with a class.

3.2. Adaptive patch split using Quad-tree decomposition

By splitting the image into small patches with fixed sizes, dehazing methods can describe the scene in detail. However, processing a lot of patches is time consuming. These methods

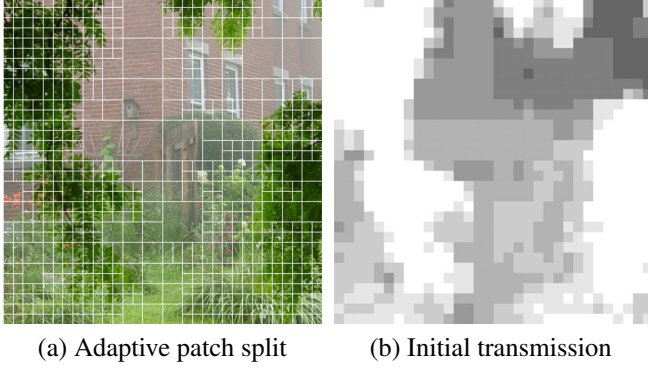


Fig. 3. Proposed quad-tree decomposition to estimate the initial transmission.

need much higher computational time when the image has higher resolution. To alleviate the aforementioned problem, our method creates adaptive patches with varying sizes by using quad-tree decomposition, as illustrated in Fig. 3(a). Unlike quad-tree searching, the quad-tree decomposition splits the image into four sub-blocks recursively. Our method splits B^d in half and obtains B^{d+1} :

$$B^{d+1} = Q_{decomp}(B^d, \lambda_1, \lambda_2, \tau), \quad (3)$$

where B^d is a sub-block in the d -level quad-tree decomposition. In (3), λ_1 and λ_2 indicate maximum (128×128) and minimum (16×16) block sizes, respectively. τ is the threshold value of 0.4. All parameters are empirically determined and fixed throughout all experiments. It should be noted that the performance of our method is not sensitive to the parameters. The decomposition terminates when it satisfies a stopping criterion that all sub-blocks are flat. We design the stopping criterion as follows:

$$Q_{decomp} = \begin{cases} \text{stop} & \text{if } \max(B^d) - \min(B^d) < \tau \\ \text{split} & \text{otherwise} \end{cases}, \quad (4)$$

where the functions $\max$ and $\min$ return the maximum and minimum intensities of the sub-block, respectively. When the difference between the two values is lower than the threshold, our method considers the sub-block flat and stops splitting it.

Adaptive patches enable our method to strike a balance between the computational cost and accuracy of patch classification. In addition, the adaptive patches meet the assumption that the transmission is locally constant. Fig. 3(b) shows patch classification results using the quad-tree decomposition.

3.3. Transmission refinement

The initial transmission discontinues at boundaries of patches, as shown in Fig. 3(b). In addition, initial transmission results can depend on the region splits or contents of the input image, which yields halos and block artifacts at the edges. Thus, we refine the initial transmission map by using guided filtering [13]. The guided filtering makes the initial

Table 1. Comparison of the number of patches according to decomposition methods.

		5×5 grid	Proposed
600×450 (Tiananmen)	Patches	10,800	136
	Run Time (sec)	87.5	1.1
440×450 (House)	Patches	7920	700
	Run Time (sec)	66.5	5.5

Table 2. Comparison with state-of-the-art dehazing methods. e : newly visible edge after recovering, σ : saturation pixel ratio, $\bar{r}$: quality of contrast restoration. Bold indicates the best result.

		House	Mountain	Tiananmen	Dolls	Fruit
MSCNN [6]	e	0.02	0.13	0.35	1.07	0.16
	σ	2.85	0.60	0.14	0	2.35
	$\bar{r}$	1.56	1.24	1.29	1.84	1.00
DehazeNet [7]	e	0.06	0.11	0.30	0.87	0
	σ	0.92	0.40	1.47	0.74	2.06
	$\bar{r}$	1.08	1.18	1.20	1.56	1.01
Proposed	e	0.09	0.11	0.49	1.62	0.17
	σ	0.13	0.49	0	0.24	0.32
	$\bar{r}$	1.52	1.41	1.49	2.63	1.06

transmission have a min channel. After the guided filtering, we use relative total variation to smooth the transmission map.

3.4. Estimating the Atmospheric light

Although atmospheric light has been estimated in many different ways, it still remains a challenging problem because saturation and color distortion can occur depending on A . In most cases, the brightest pixel is selected as A . He *et al.* select A among the brightest pixels in the dark channel. The quad-tree searching method [9] chooses A by recursively dividing the image into four sub-blocks and entering the sub-block which has the highest average intensity. Kim and Kwon [14] propose pixel-wise atmospheric light estimation, in which the illumination of each pixel is used as the atmospheric light based on the Retinex theory.

Unlike these methods, our method uses patch classification results for the atmospheric light estimation. The method chooses pixels with the lowest 0.001% intensity in the initial transmission map such as Fig. 3(b). Then, the method determines A as the brightest value in the R, G, or B channel at the corresponding pixels of the input image.

4. EXPERIMENTS

Our method was compared with other state-of-the-art deep learning based dehazing algorithms [6, 7] both quantitatively and qualitatively. We used VLD [16] and FADE [15] as quantitative evaluation metrics. Our method adopted VGGNet for patch classification. About half of our 100,000 synthetic data

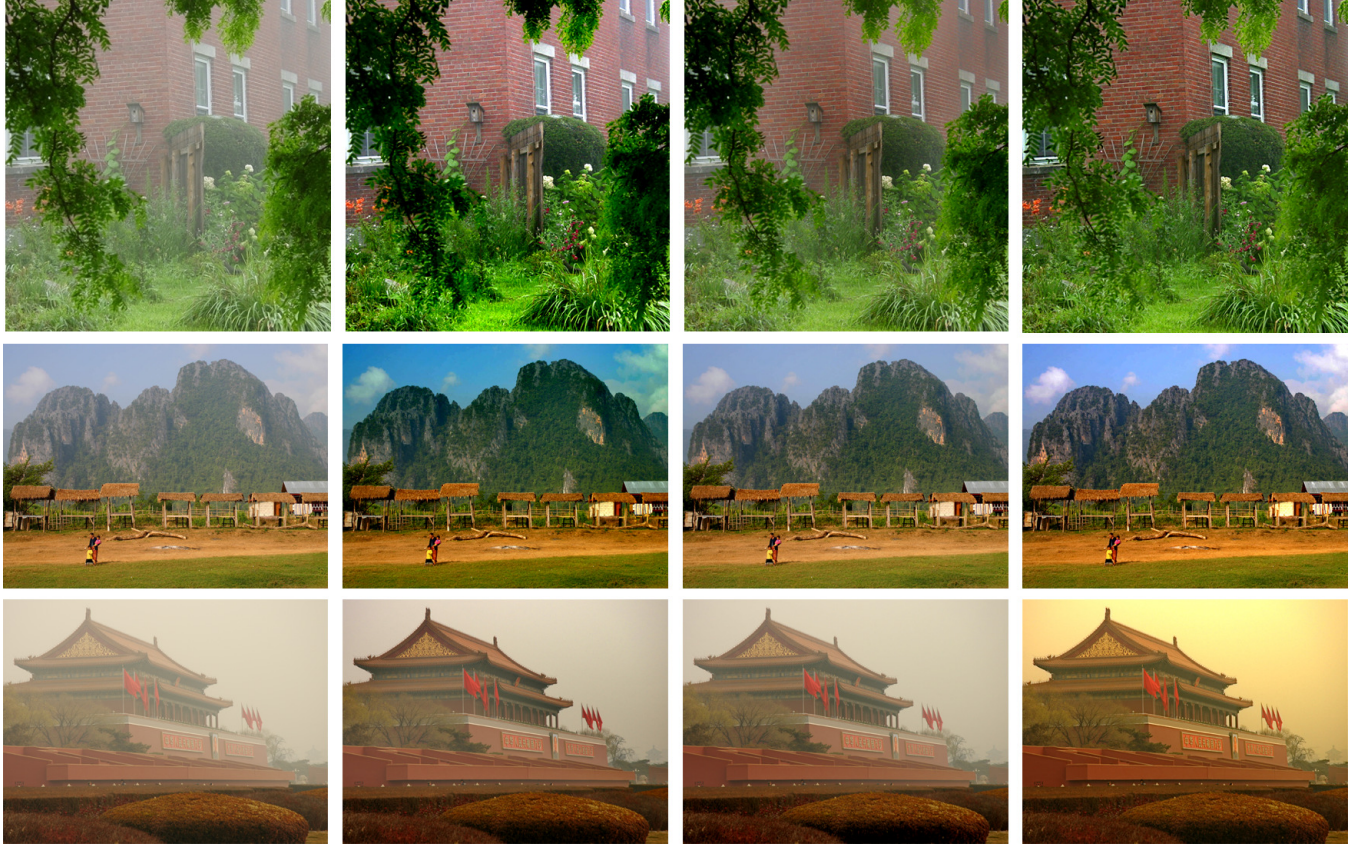


Fig. 4. Dehazing results in House, Mountain and Tiananmen. Left to Right: Input image, MSCNN [6], DehazeNet [7], and the proposed method.

Table 3. Perceptual Fog Density measure (FADE) in [15]. The lower the value is, the better.

	MSCNN [6]	DehazeNet [7]	Proposed
House	0.883	1.224	1.142
Mountain	1.243	1.561	1.357
Tiananmen	1.842	1.704	1.474

were used for training, and the other half for the validation test. We adopted cross entropy as a loss function and used a mini- batch size of 64 with the Adam optimizer. The model was trained for a total of 10 epochs with a learning rate of $1e^{-5}$. We conducted the experiments on a workstation with an NVIDIA GeForce GTX TITAN XP GPU.

Table 1 reports the number of patches and the corresponding classification time for a regular grid of 5×5 and for our adaptive grid using quad-tree decomposition. Notably, the more an input image contains homogeneous regions, the bigger the gap is between the two approaches even in the same resolution. Also, it makes a huge difference in the computation time between them.

In Table 2, e is the newly visible edge ratio, σ the saturated pixel ratio and $\bar{r}$ the quality of contrast restoration. As shown in the table, the proposed method is the best in terms of e , $\bar{r}$, and σ . In Table 3, FADE predicts the local perceptual density of each patch and also provides a perceptual density score

of the entire image. MSCNN produced good results on the House, in terms of $\bar{r}$ and the FADE score, because the House is a high contrast image.

As demonstrated in Fig. 4, MSCNN and DehazeNet lost local details of the wood parts of the House. In addition, a halo effect occurred near the edges in the case of DehazeNet. For the Mountain and Tiananmen images, our haze-free results were clearer than for the other algorithms.

5. CONCLUSION

In this paper, we propose a patch-classification based dehazing method with adaptive patch splits. To estimate the transmission map, we train a deep learning network with synthetic data of hazy and haze-free pairs. The network classifies patches with adaptive sizes, in which the size is determined by using quad-tree decomposition. With adaptive patches, our method significantly reduces the number of patches compared with conventional approaches. In experiments, our method outperforms state-of-the-art dehazing methods both quantitatively and qualitatively.

6. ACKNOWLEDGMENT

This work was supported by Institute for Information & communications Technology Promotion(IITP) grant funded by the Korea government(MSIP) (No.2017-0-01780).

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